



DATA INSIGHTS FOR ESAF BANK'S STRATEGIC GROWTH IN INDIA

06/07/2025

PREPARED BY:

Mahima Johnson

2ND Year Bsc Data Science & Statistics

Christ University, Central Campus,
Bangalore

Acknowledgement

I would like to express my sincere gratitude to all those who contributed to the successful completion of this project.

First and foremost, I am deeply thankful to **Ms. Mala Nair** for her invaluable support and guidance throughout this project. Her insightful suggestions and the idea behind this project played a key role in shaping its direction and purpose.

I express my appreciation to **Ms. Sandhya**, to whom I am submitting this project. Thank you for your time, consideration, and for evaluating my work.

Finally, I would also like to extend my heartfelt thanks to **Ms. Reji Koshy** for giving me the opportunity to undertake and present this work. Her encouragement and trust in my abilities have been truly motivating.

This project would not have been possible without their encouragement, mentorship, and support.

Table of Content

01	Introduction
02	Objectives
03	Data Description
04	Steps Involved
05	Tools Used
06	Visualization & Interpretation
07	Conclusion

Introduction

This project aims to identify the possible locations ESAF Small Finance bank can set up their branches. The bank targets urban and semi urban places areas in order to accomodate financial inclusions. Using a data-driven approach, this project explores various socio-economic and regional indicators to assist in strategic expansion planning.

Objectives: -

The objectives of the project is as follows: -

- To analyze and interpret data related to geographic and financial behavior across the different states in India.
- To visualize and compare potential locations using clustering and geographic insights.
- To support business decision-making with reproducible, code-based analysis.

Data Description

The analysis in this project is based on three datasets that provide crucial socio-economic indicators relevant to understanding financial literacy and banking access across Indian states. The datasets are sourced from **official government portals and credible publications**, ensuring reliability and validity of the insights derived.

1. State-wise Population Dataset (Population_States.csv)

- **Source:** Reserve Bank of India (RBI) website
- **Description:**
This dataset presents the total population of each Indian state across different census years, ranging from **1951 to 2011**.
- **Variables Include:**
 - State: Name of the state or union territory.
 - 1951 to 2011: Population values recorded in crores for each decadal census year.
- **Usage in Project:**
 - Used to analyze population growth trends across states.
 - Helpful in understanding population pressure on financial and banking services.

2. Financial Literacy Rate Dataset (financial_literacy_rate.pdf)

- **Source:**

Extracted from the research paper titled "Inter-State Variation of Financial Literacy in India" published in the **International Journal of Creative Research Thoughts (IJCRT)**, Volume 11, Issue 11, 2023.

Data is based on **Kantar's Financial Inclusion Insights (FII) Wave 6 Survey (2018)**.

- **Description:**

This dataset presents **state-wise financial literacy rates** and scores across three components:

- **Financial Knowledge**
- **Financial Behavior**
- **Financial Attitude**

- **Measurement Framework:**

- Based on **OECD-INFE standards**, financial literacy is a composite score (0–16 scale) derived from performance in all three components.
- Adults scoring **10 or above** are considered financially literate.

- **Coverage:**

- 28 States and UTs of India, with data representing adults aged **18 years and above**.
- Over **48,000 respondents**, surveyed using stratified multistage sampling.

- **Usage in Project:**

- Enables comparison of financial literacy levels across states.
- Assists in correlating financial literacy with population size and rural distribution.
- Identifies states needing focused financial education interventions.

3. Rural Population Dataset (rural_pop.csv)

- **Source:** Reserve Bank of India (RBI) website
- **Description:**
This dataset captures the **rural population distribution across Indian states** during the same census years (1951 to 2011).
- **Variables Include:**
 - State: State/UT name.
 - 1951 to 2011: Rural population values corresponding to each census year.
- **Usage in Project:**
 - Helps identify the proportion of rural population in each state.
 - Critical in evaluating **banking inclusion in rural areas** and aligning it with financial literacy levels.
 - Supports rural-vs-urban comparisons in financial services access.

Steps Involved

The project involves the following steps:

01 Data Collection

02 Data Cleaning and Preprocessing

03 Exploratory Data Analysis

04 Clustering and Modeling

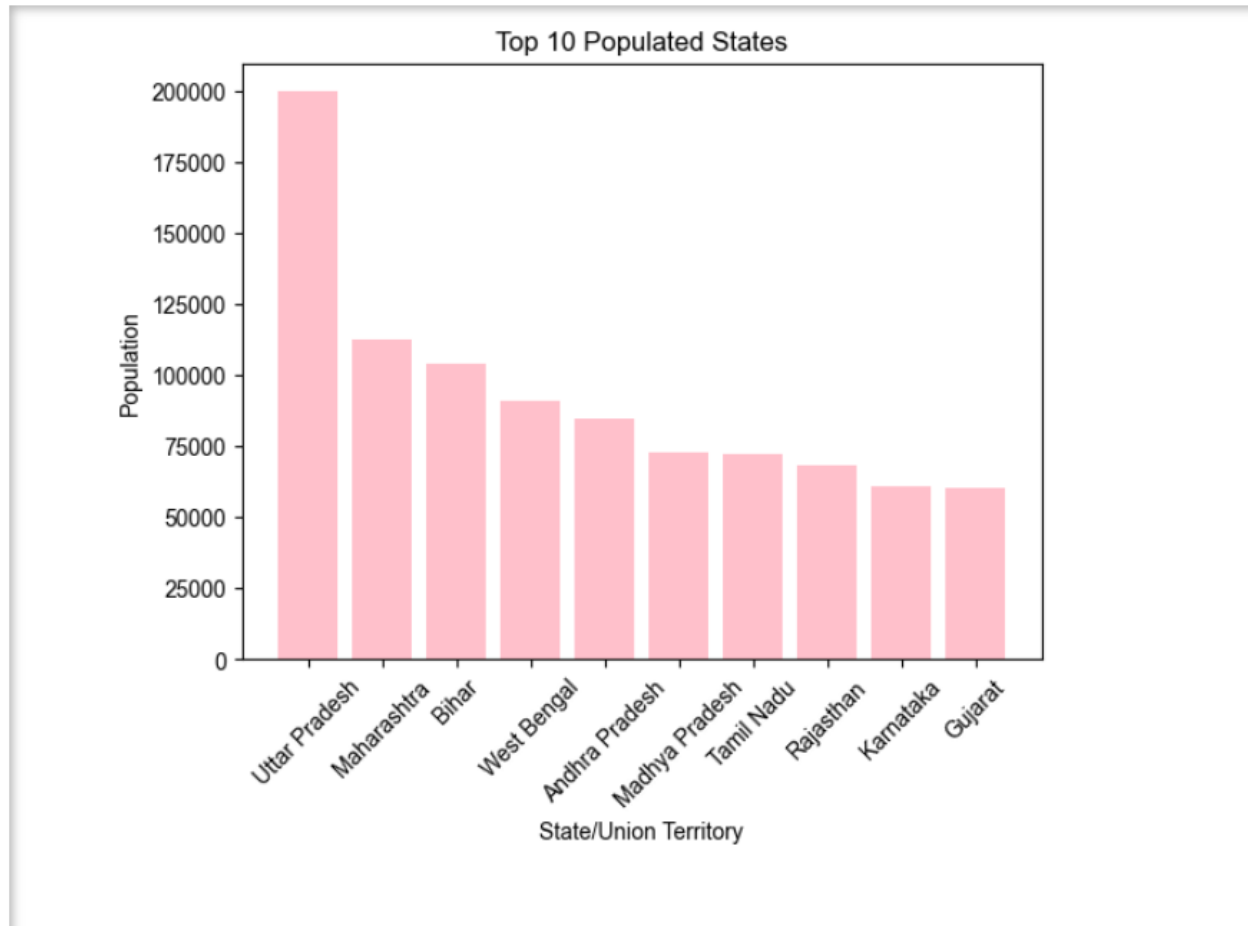
05 Insights and Recommendations

Tools Used

- **Python**- Core Programming Language
- **Jupyter Notebook** -Coding Environment
- **PANDAS** Data Handling and Preprocessing
- **MATPLOTLIB** Data Visualization
- **Scikit-learn** Clustering and modeling

Visualization and Interpretation

01 Top 10 Populated States.

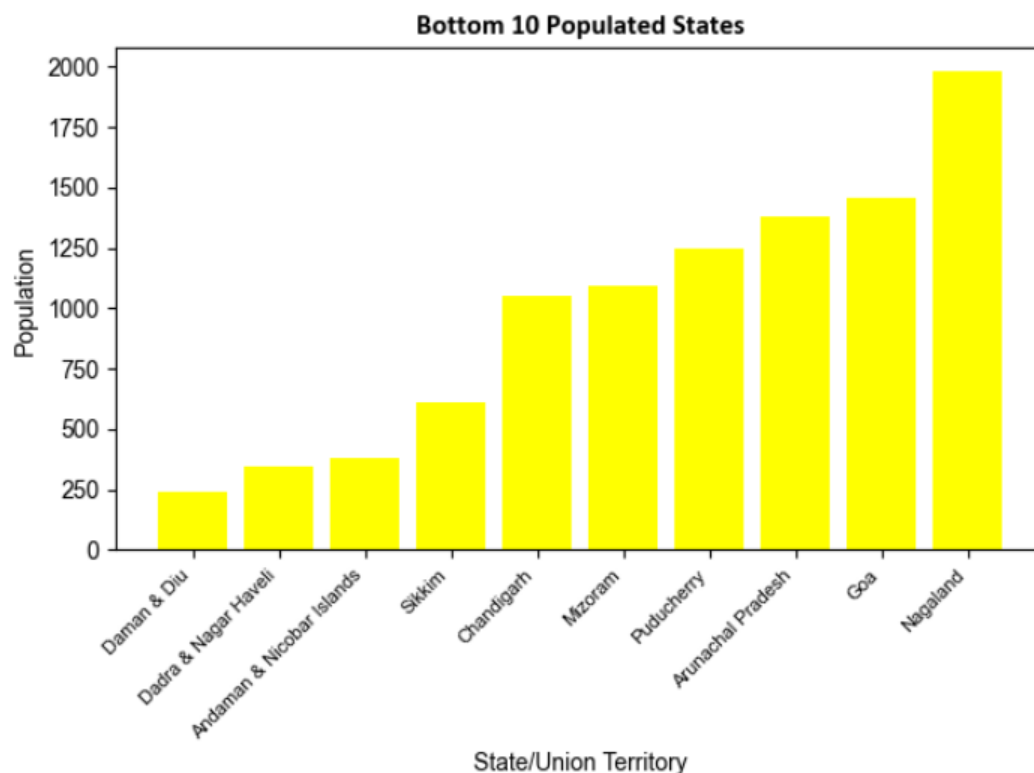


Description:

This bar chart shows the top 10 most populated states/UTs in India. Uttar Pradesh has the highest population, followed by Maharashtra, Bihar, and West Bengal.

Interpretation:

Uttar Pradesh stands out with a significantly higher population than all other states, indicating a dense demographic concentration. States like Maharashtra and Bihar also have high populations, which implies higher demand for financial services, infrastructure, and literacy initiatives. These states are key targets for banking expansion and financial inclusion strategies due to the large consumer base.

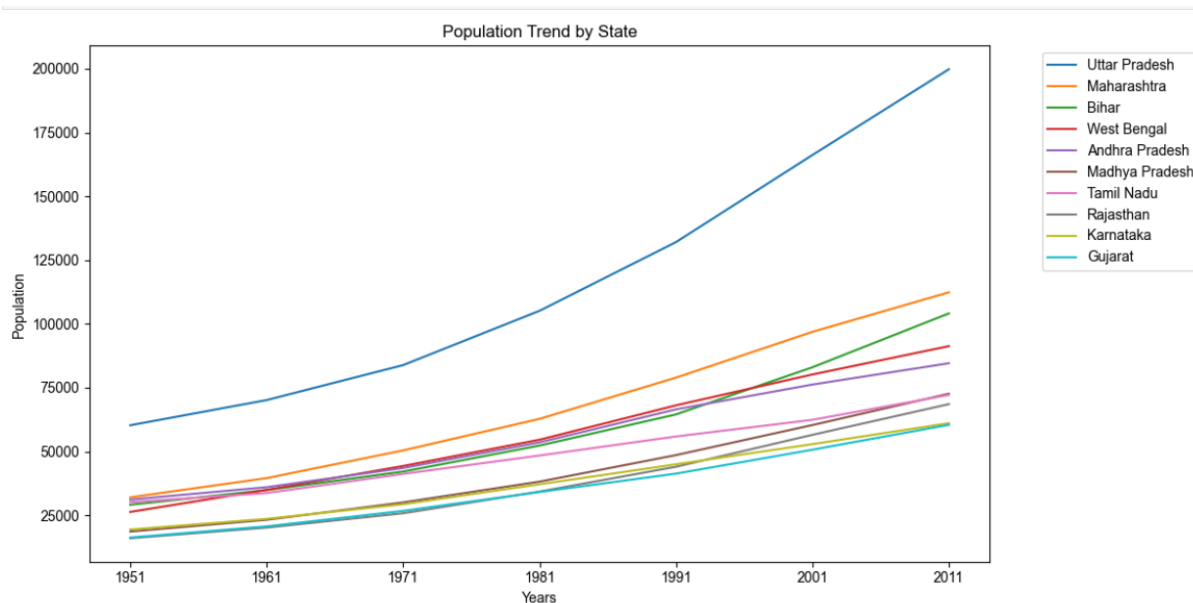
02 Bottom 10 Populated States

Description:

This chart shows the 10 states/UTs with the lowest population figures. Daman & Diu and Dadra & Nagar Haveli have the smallest populations, while Nagaland tops the list among the lowest 10.

Interpretation:

Smaller states and union territories like Daman & Diu and Andaman & Nicobar Islands have very limited populations. While financial outreach may be easier due to scale, geographical and infrastructural challenges can still hinder access. These regions require more personalized and region-specific financial awareness programs and digital banking initiatives.

03 Population Trend (Top Populated States – Line Chart)**Description:**

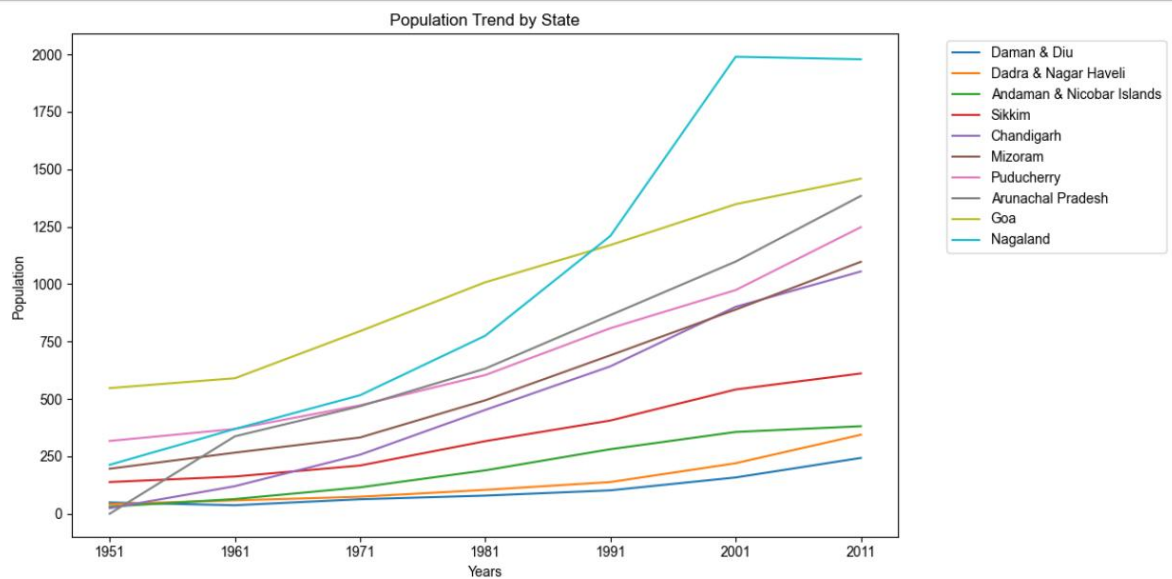
This line graph represents the population growth of top populated states from 1951 to 2011. Each line traces how the population has changed across the decades.

Interpretation:

All top 10 populated states show consistent upward growth, with Uttar Pradesh maintaining a significant lead. The rapid increase in states like Bihar and

Maharashtra highlights the growing demand for financial products, literacy, and banking infrastructure. The trends underline the need for sustainable planning in banking and economic development in these regions.

04 Population Trend (Bottom Populated States – Line Chart)



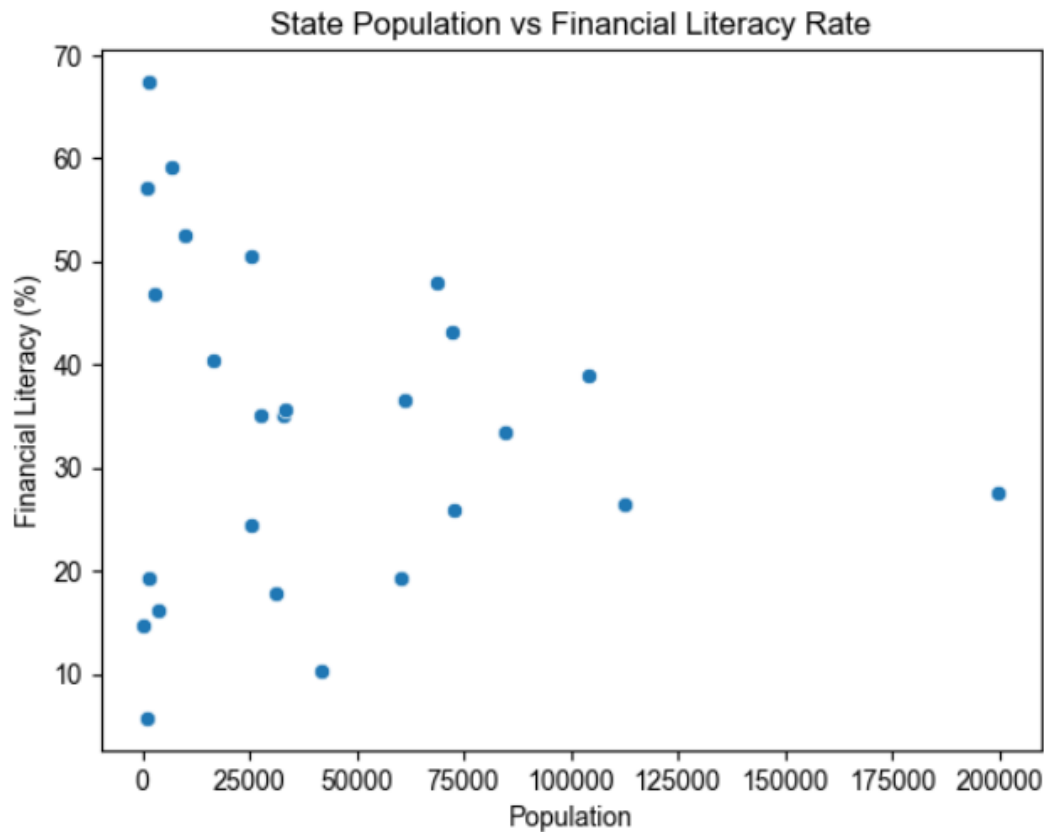
Description:

This chart shows population growth in the 10 least populated states and union territories over time from 1951 to 2011.

Interpretation:

Despite their small size, these states show steady growth over time, particularly Nagaland and Goa. Some UTs like Chandigarh have shown notable population increases due to urban migration. These areas may have limited but growing demand for banking services. The focus here should be on accessibility—mobile banking, awareness, and rural outreach may play a larger role.

05 State Population vs Financial Literacy Rate



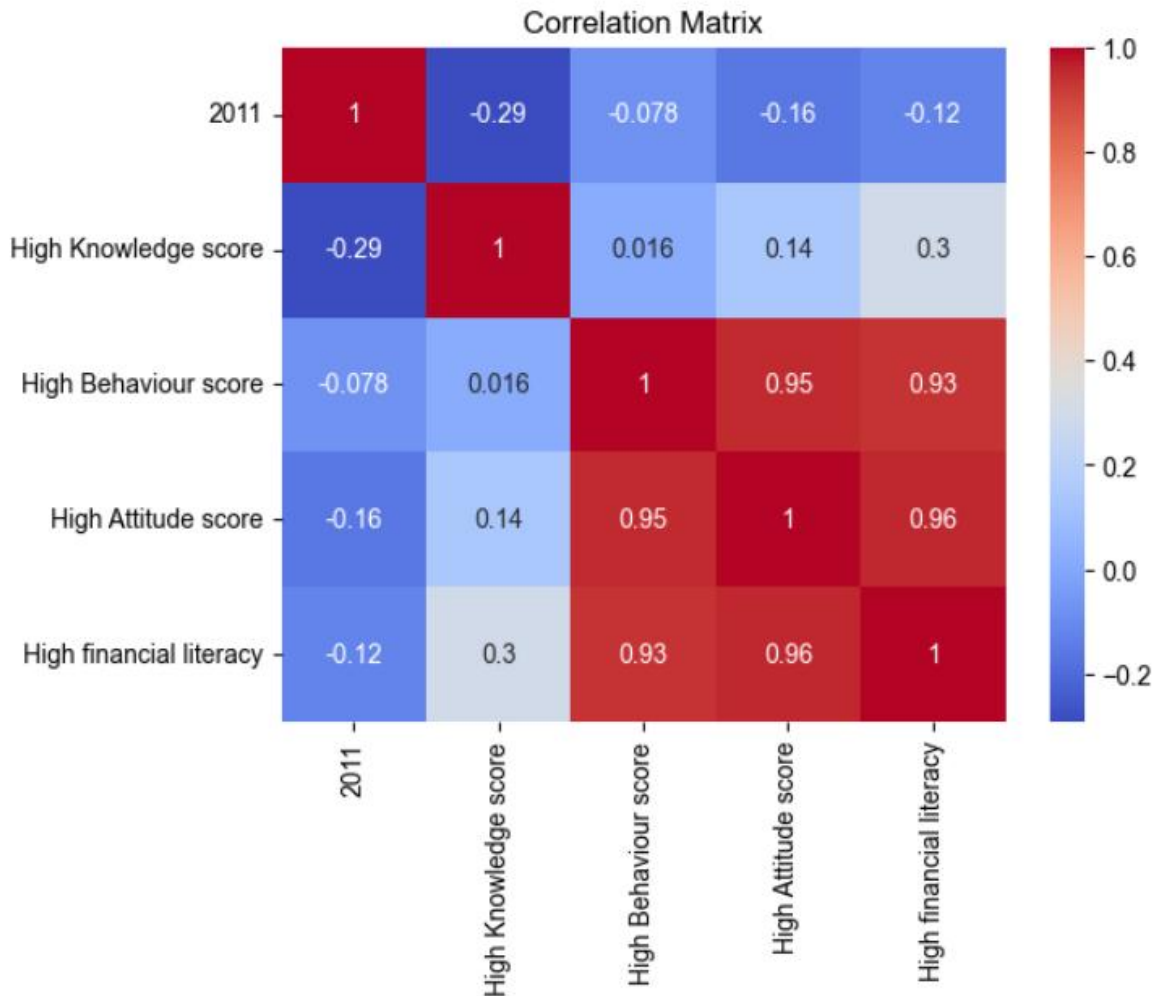
Description:

This scatter plot plots every Indian state as a single dot. The horizontal axis shows the total population of the state, while the vertical axis shows the percentage of adults judged to be financially literate. Populations span from a few thousand to roughly two-hundred million people, and literacy rates range from about 5 percent up to nearly 70 percent.

Interpretation:

The cloud of points slopes gently downward, which tells us that bigger states tend to report lower financial-literacy scores. In other words, the more people a state has, the harder it seems to be to achieve high levels of financial knowledge and skills across the whole population. This pattern hints that densely populated regions may need extra-large, targeted literacy campaigns to reach everyone effectively.

06 Core Financial-Capability Correlation Matrix



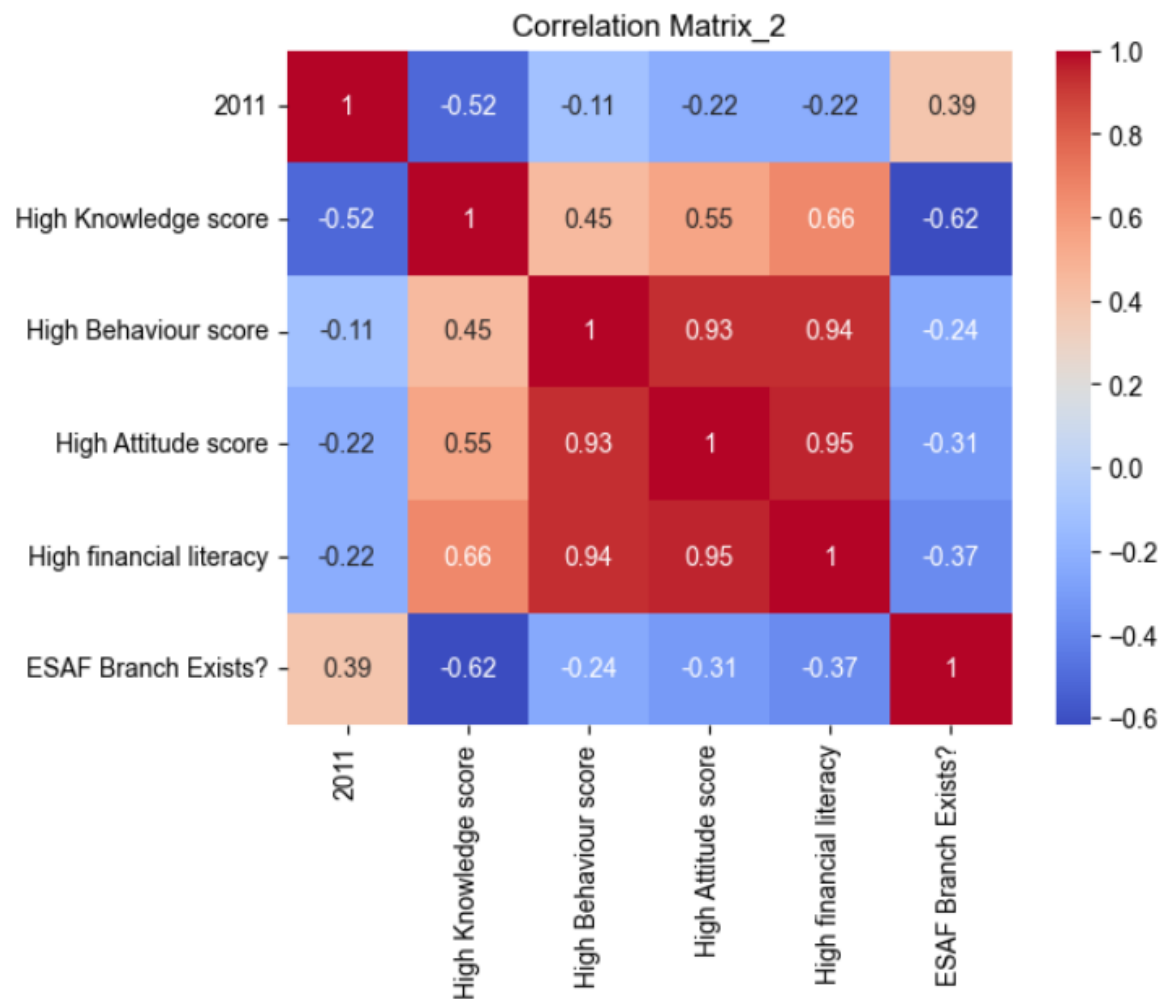
Description:

The first heat-map presents a square matrix of correlation coefficients among four survey constructs—High Knowledge Score, High Behaviour Score, High Attitude Score and overall High Financial Literacy. Dark red cells signal strong positive relationships, whereas dark blue cells indicate strong negative ones.

Interpretation:

High Behaviour and High Attitude are almost perfectly aligned with High Financial Literacy, showing that people who practise good money habits and hold positive mind-sets about finance are also the ones who score best on overall literacy. Knowledge is helpful but less decisive, suggesting that factual know-how alone is not enough; attitudes and day-to-day behaviours truly drive financially sound decisions.

07 Extended Correlation Matrix with “ESAF Branch Exists?”



Description:

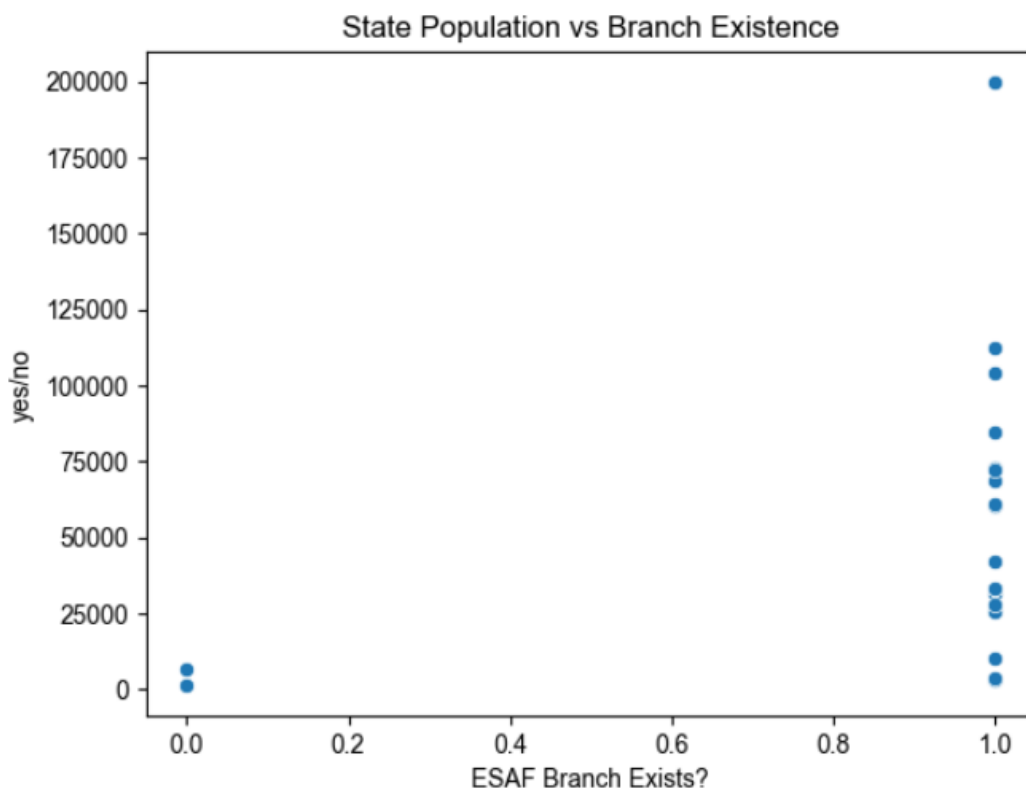
A second heat-map repeats the earlier variables and adds a binary flag that marks whether a state hosts at least one ESAF Small-Finance Bank branch. The same colour scale illustrates how the presence of a branch relates to knowledge, behaviour, attitude and literacy.

Interpretation:

The ESAF-branch column is moderately negative against knowledge, behaviour, attitude and literacy, meaning ESAF tends to open in places where all four capability measures are weaker. That pattern supports the bank’s stated mission

of serving financially underserved communities: branches appear where the need for improved literacy and inclusion is greatest.

08 ESAF Branch Presence vs State Population



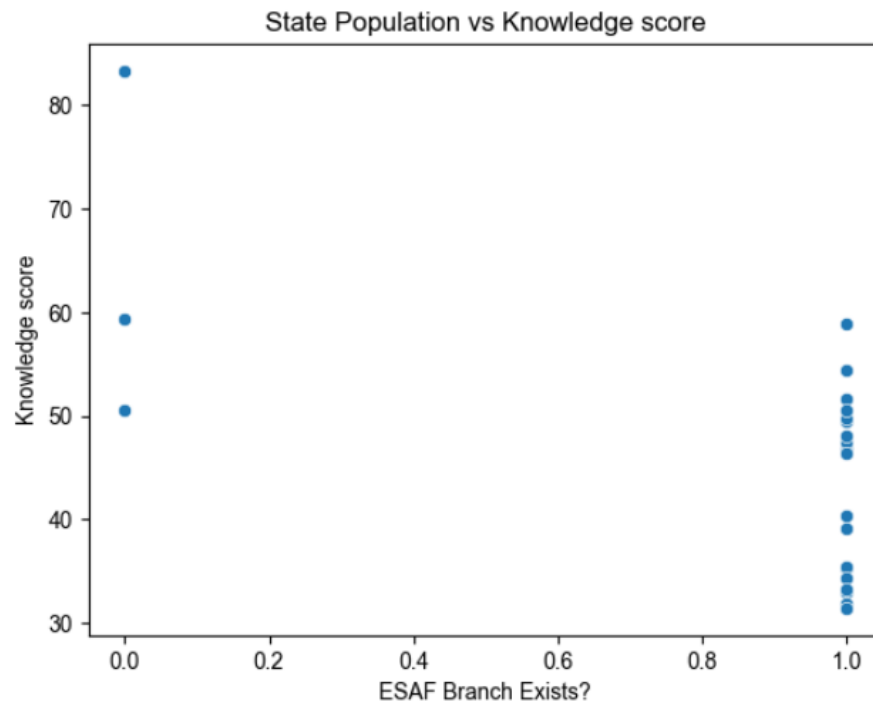
Description:

This scatter plot places every state into one of two vertical bands—"0" for states with no ESAF branch and "1" for those with at least one. The vertical axis represents the state's total population.

Interpretation:

Almost all high-population states sit in the "branch present" band, while the few without branches are low-population territories. ESAF therefore concentrates on larger markets, presumably because a single branch can serve many more potential customers and amplify the bank's social-impact reach.

09 ESAF Branch Presence vs High Knowledge Score



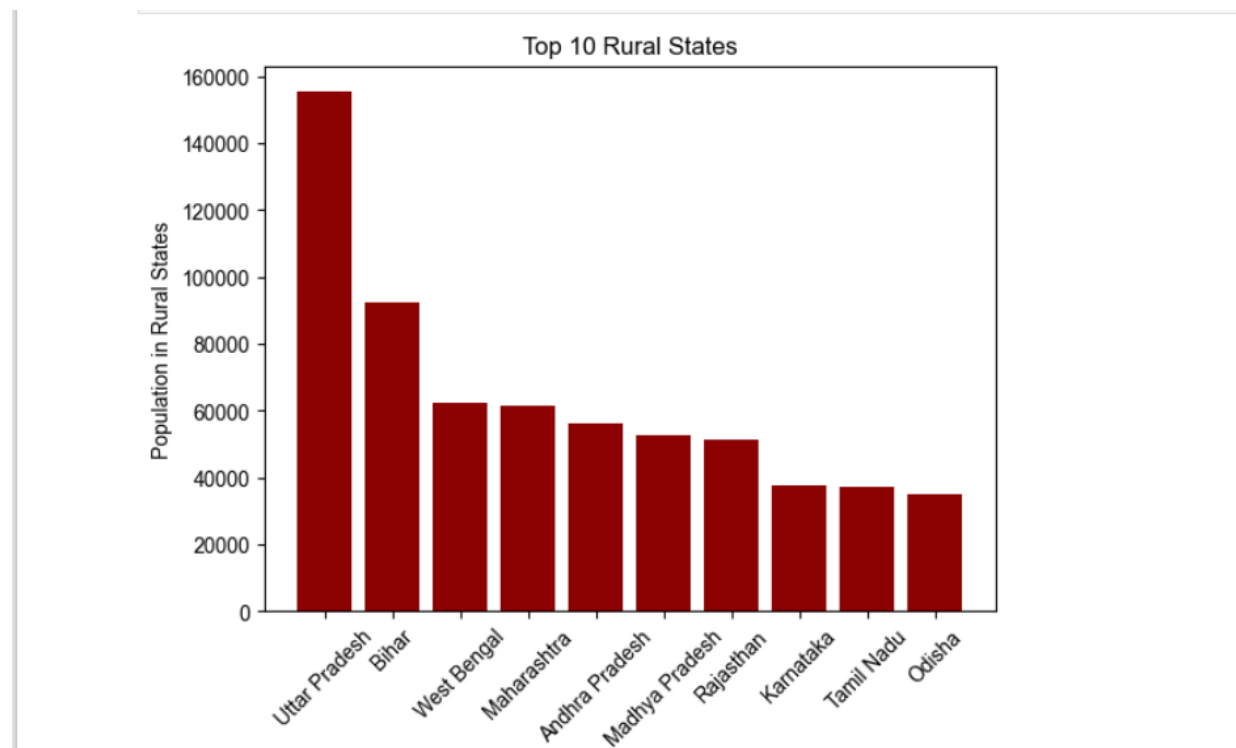
Description:

Here the horizontal axis again indicates whether a branch exists (0 or 1), but the vertical axis now shows each state's High Knowledge Score. Each dot summarises one state.

Interpretation:

States with ESAF branches span a wide range of knowledge outcomes, from low to high, whereas the few states lacking a branch cluster at the middle-to-higher end. Combined with the earlier correlation map, this suggests ESAF does not shy away from low-knowledge environments; instead, it may deliberately choose them to help close capability gaps.

10 Top Ten Rural States by Population



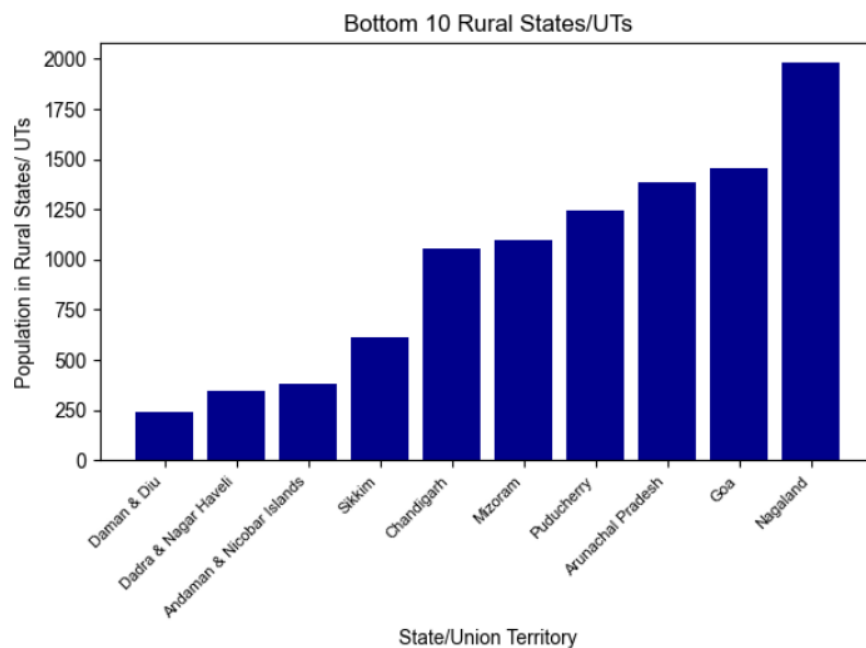
Description:

This bar chart ranks the ten largest rural populations in 2011. Uttar Pradesh tops the list by a wide margin, followed by Bihar, West Bengal and Maharashtra; the remaining bars represent Andhra Pradesh, Madhya Pradesh, Rajasthan, Karnataka, Tamil Nadu and Odisha.

Interpretation:

Because these states house the majority of India's rural residents, they present the greatest opportunities—and challenges—for rural banking, micro-credit and literacy drives. Any nationwide initiative that overlooks these ten regions would fail to reach most rural households.

11 Bottom Ten Rural States/UTs by Population



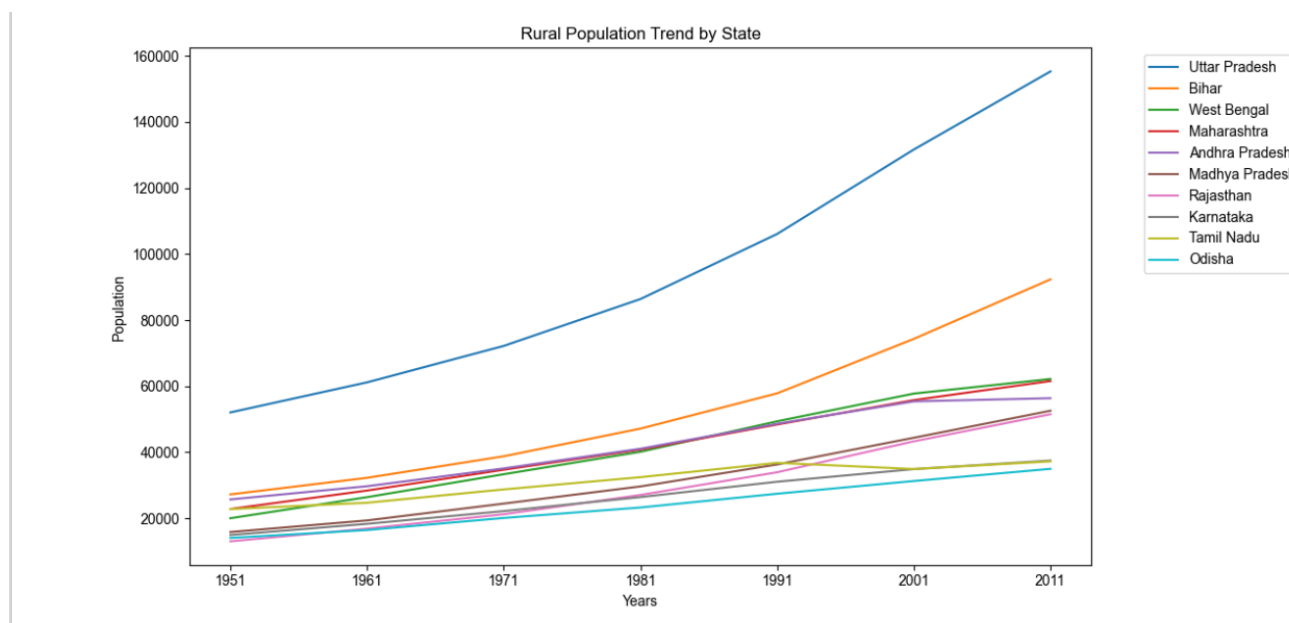
Description:

A second bar chart displays the rural population of the ten smallest states or union territories, beginning with Daman & Diu and extending up to Nagaland. The tallest bar in this chart is still under two million inhabitants.

Interpretation:

These territories are comparatively tiny markets. Physical branch expansion may yield limited returns here, but lightweight, digital-first solutions could still deliver financial services efficiently without the heavy fixed costs of extensive branch networks.

12 Rural Population Trend by State, 1951 – 2011



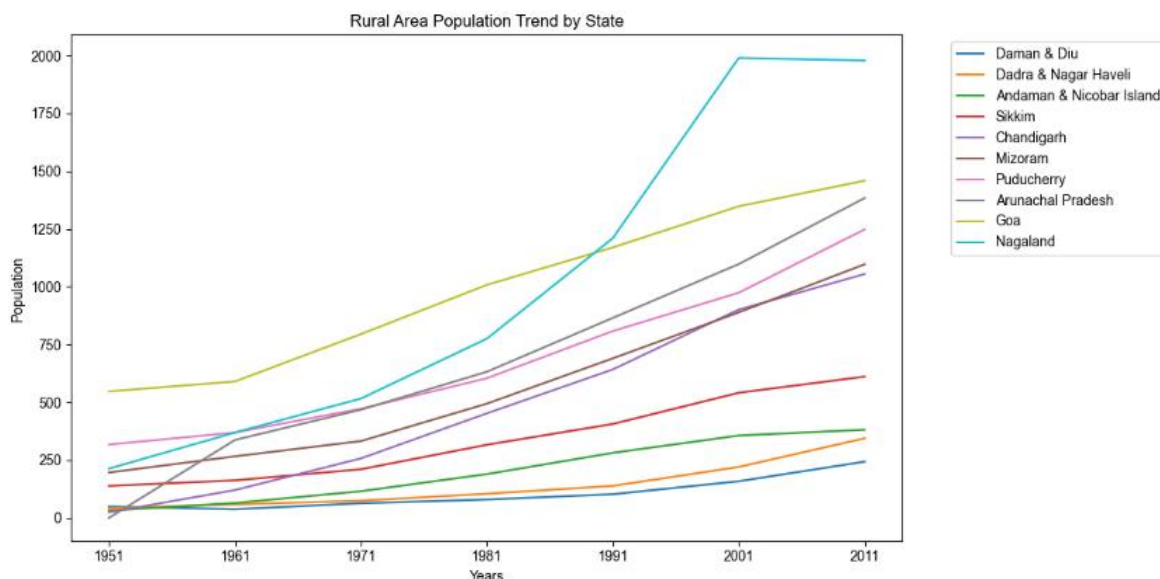
Description:

The line graph tracks rural-population growth in the ten largest states at decennial census points from 1951 to 2011. Each coloured line represents one state, and all lines trend upward over time.

Interpretation:

Every state shows sustained rural-population growth, but the pace differs sharply. Uttar Pradesh accelerates the most, while Tamil Nadu and Odisha grow more modestly. For banks, the takeaway is clear: demand for rural financial services keeps rising, and growth hotspots such as Uttar Pradesh and Bihar will require continuous scaling of both infrastructure and literacy programs to keep pace with demographic pressure.

13 Rural Area Population Trend by State (Bottom 10 Populated States)



Description:

This line graph displays the rural population trends from 1951 to 2011 for the 10 least populated states and union territories, including Nagaland, Goa, Mizoram, and others.

Interpretation:

The chart reveals a **steady and consistent increase in rural population** across all listed regions over the decades. Notably:

- **Nagaland** shows the sharpest rural growth, especially from 1981 to 2001, indicating either natural growth or improvements in rural data collection and migration from urban to rural fringes.
- **Goa** and **Arunachal Pradesh** also display continuous rural growth, suggesting rising residential development or rural retention.
- **Daman & Diu** and **Dadra & Nagar Haveli** remain at the lower end, with a much smaller rural population, likely due to their limited geographical area and industrial focus rather than agrarian settlements.
- **Chandigarh**, while traditionally considered urban, still shows rural figures—possibly due to peri-urban expansion or reclassification.

Insight for Banking & Financial Services:

Even small states and UTs are showing rural population growth. This emphasizes the need for **rural financial inclusion**, especially in remote or tribal regions like Mizoram, Nagaland, and Arunachal Pradesh. Financial institutions, especially small finance banks and microfinance agencies, can leverage this trend by offering **mobile banking, self-help group loans, or agricultural credit schemes** tailored for rural populations.

Conclusion

The population analysis across Indian states reveals critical insights into demographic concentration and growth trends that have direct implications for financial services and banking outreach. States like **Uttar Pradesh, Maharashtra, and Bihar** continue to dominate in terms of overall population, indicating a significant demand for banking infrastructure, financial literacy programs, and credit facilities to meet the needs of their large and diverse populations.

Conversely, smaller states and Union Territories such as **Daman & Diu, Sikkim, and Nagaland**, while less populated, show consistent population growth—particularly in **rural areas**. This rural demographic expansion highlights the importance of enhancing **financial inclusion** in remote and underserved regions through tailored services like mobile banking, microfinance, and rural credit delivery models.

The trends suggest that a one-size-fits-all approach to financial planning will not be effective. Instead, banking institutions must **align their strategies** with both population scale and growth patterns. Larger states require **broad-based infrastructure and service coverage**, whereas smaller states may benefit from **targeted, technology-driven outreach programs**.

Overall, understanding population dynamics is crucial for **optimizing banking penetration, improving financial literacy, and promoting inclusive growth** across India.